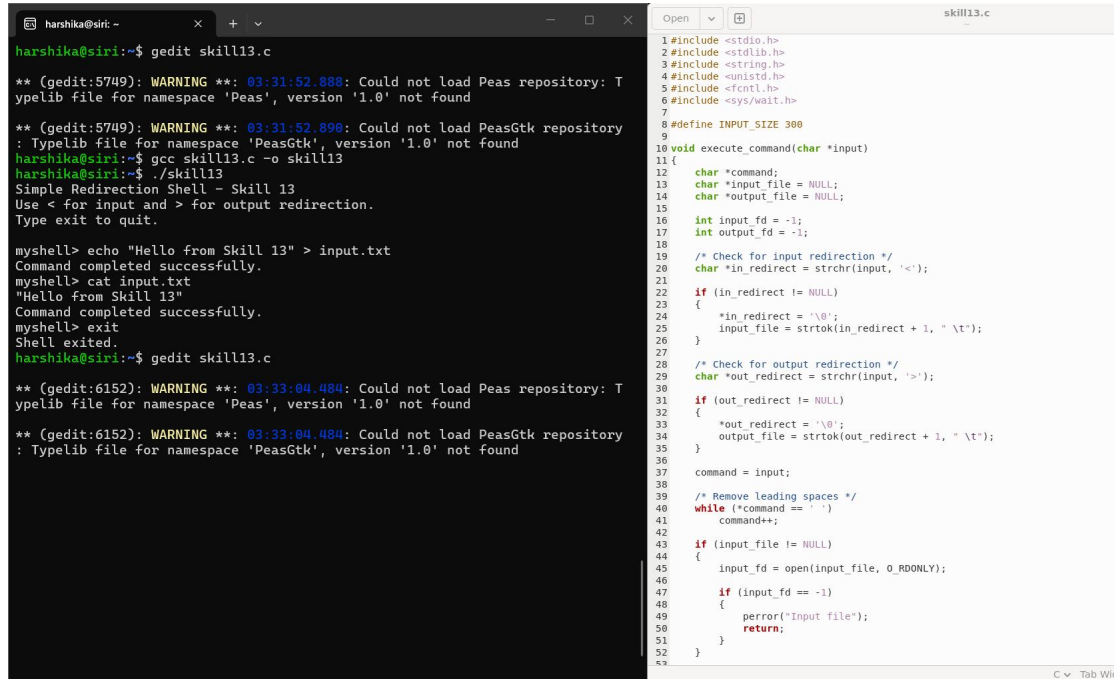


Skill - 13

1. Input and Output redirection



The image shows a terminal window on the left and a code editor on the right. The terminal window, titled 'harshika@siri: ~', shows the user creating a file 'skill13.c' with 'gedit'. It then displays several warning messages from 'Peas' (Peas repository and PeasGtk repository not found). The user then runs 'gcc skill13.c -o skill13' and './skill13'. The program output is 'Simple Redirection Shell - Skill 13' and 'Use < for input and > for output redirection. Type exit to quit.' The user then runs 'myshell> echo "Hello from Skill 13" > input.txt', 'myshell> cat input.txt' (outputting 'Hello from Skill 13'), and 'myshell> exit'.

```
harshika@siri:~$ gedit skill13.c

** (gedit:5749): WARNING **: 03:31:52.888: Could not load Peas repository: TypeLib file for namespace 'Peas', version '1.0' not found

** (gedit:5749): WARNING **: 03:31:52.890: Could not load PeasGtk repository: TypeLib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc skill13.c -o skill13
harshika@siri:~$ ./skill13
Simple Redirection Shell - Skill 13
Use < for input and > for output redirection.
Type exit to quit.

myshell> echo "Hello from Skill 13" > input.txt
Command completed successfully.
myshell> cat input.txt
"Hello from Skill 13"
Command completed successfully.
myshell> exit
Shell exited.
harshika@siri:~$ gedit skill13.c

** (gedit:6152): WARNING **: 03:33:04.484: Could not load Peas repository: TypeLib file for namespace 'Peas', version '1.0' not found

** (gedit:6152): WARNING **: 03:33:04.484: Could not load PeasGtk repository: TypeLib file for namespace 'PeasGtk', version '1.0' not found
```

```
1#include <stdio.h>
2#include <stdlib.h>
3#include <string.h>
4#include <unistd.h>
5#include <fcntl.h>
6#include <sys/wait.h>
7
8#define INPUT_SIZE 300
9
10void execute_command(char *input)
11{
12    char *command;
13    char *input_file = NULL;
14    char *output_file = NULL;
15
16    int input_fd = -1;
17    int output_fd = -1;
18
19    /* Check for input redirection */
20    char *in_redirect = strchr(input, '<');
21
22    if (in_redirect != NULL)
23    {
24        *in_redirect = '\0';
25        input_file = strtok(in_redirect + 1, " \t");
26    }
27
28    /* Check for output redirection */
29    char *out_redirect = strchr(input, '>');
30
31    if (out_redirect != NULL)
32    {
33        *out_redirect = '\0';
34        output_file = strtok(out_redirect + 1, " \t");
35    }
36
37    command = input;
38
39    /* Remove leading spaces */
40    while (*command == ' ')
41        command++;
42
43    if (input_file != NULL)
44    {
45        input_fd = open(input_file, O_RDONLY);
46
47        if (input_fd == -1)
48        {
49            perror("Input file");
50            return;
51        }
52    }
```